5.1.2 Excretion as an example of homeostatic control

The kidneys, liver and lungs are all involved in the removal of toxic products of metabolism from the blood and therefore contribute to homeostasis. The kidneys play a major role in the control of the water potential of the blood.

The liver also metabolises some toxins that are ingested.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the term *excretion* and its importance in maintaining metabolism and homeostasis	To include reference to the importance of removing metabolic wastes, including carbon dioxide and nitrogenous waste, from the body.
(b)	**(i)** the structure and functions of the mammalian liver **(ii)** the examination and drawing of stained sections to show the histology of liver tissue	To include the gross structure and histology of the liver **AND** the roles of the liver in: • storage of glycogen • detoxification • the formation of urea from ammonia reacting with carbon dioxide as part of the ornithine cycle. Learners are **not** required to know details of the ornithine cycle. **PAG1** HSW4
(c)	**(i)** the structure, mechanisms of action and functions of the mammalian kidney **(ii)** the dissection, examination and drawing of the external and internal structure of the kidney **(iii)** the examination and drawing of stained sections to show the histology of nephrons	To include the gross structure and histology of the kidney including the detailed structure of a nephron and its associated blood vessels **AND** the processes of ultrafiltration, selective reabsorption and the production of urine. *M0.1, M0.3, M1.1, M1.3, M2.1, M3.1* **PAG1, PAG2** HSW4, HSW6, HSW8
(d)	the control of the water potential of the blood	To include the role of osmoreceptors in the hypothalamus, the posterior pituitary gland, ADH and its effect on the walls of the collecting ducts. HSW8
(e)	the effects of kidney failure and its potential treatments	To include the problems that arise from kidney failure including the effect on glomerular filtration rate (GFR) and electrolyte balance **AND** the use of renal dialysis (haemodialysis only) and transplants for the treatment of kidney failure. HSW7, HSW9, HSW12

(f)	how excretory products can be used in medical diagnosis.	To include the use of urine samples in diagnostic tests, with reference to the use of monoclonal antibodies in pregnancy testing and testing for anabolic steroids and drugs. **PAG9** HSW7, HSW9, HSW11, HSW12